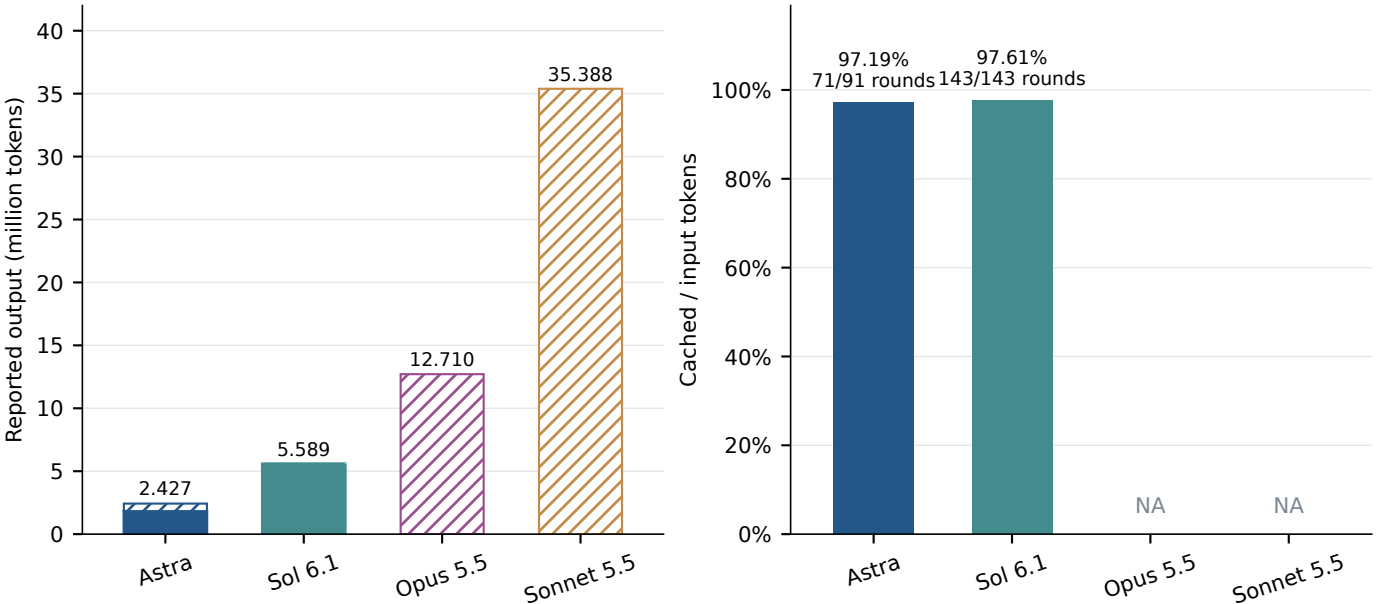
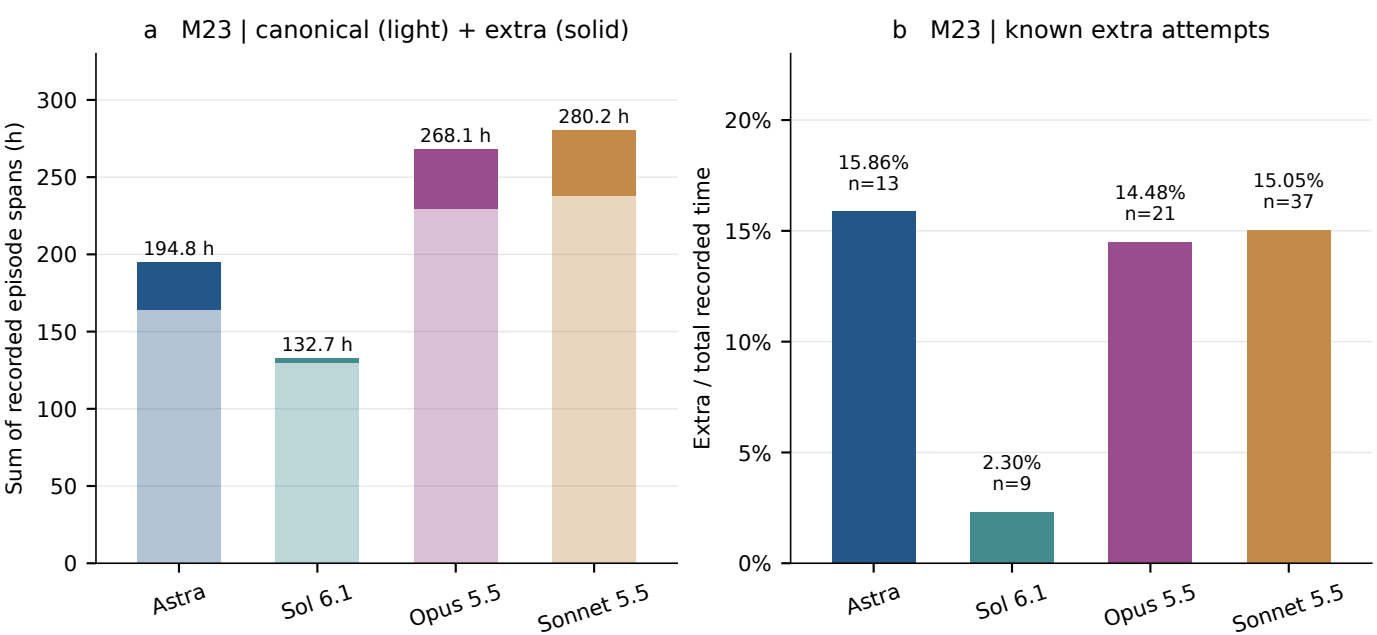
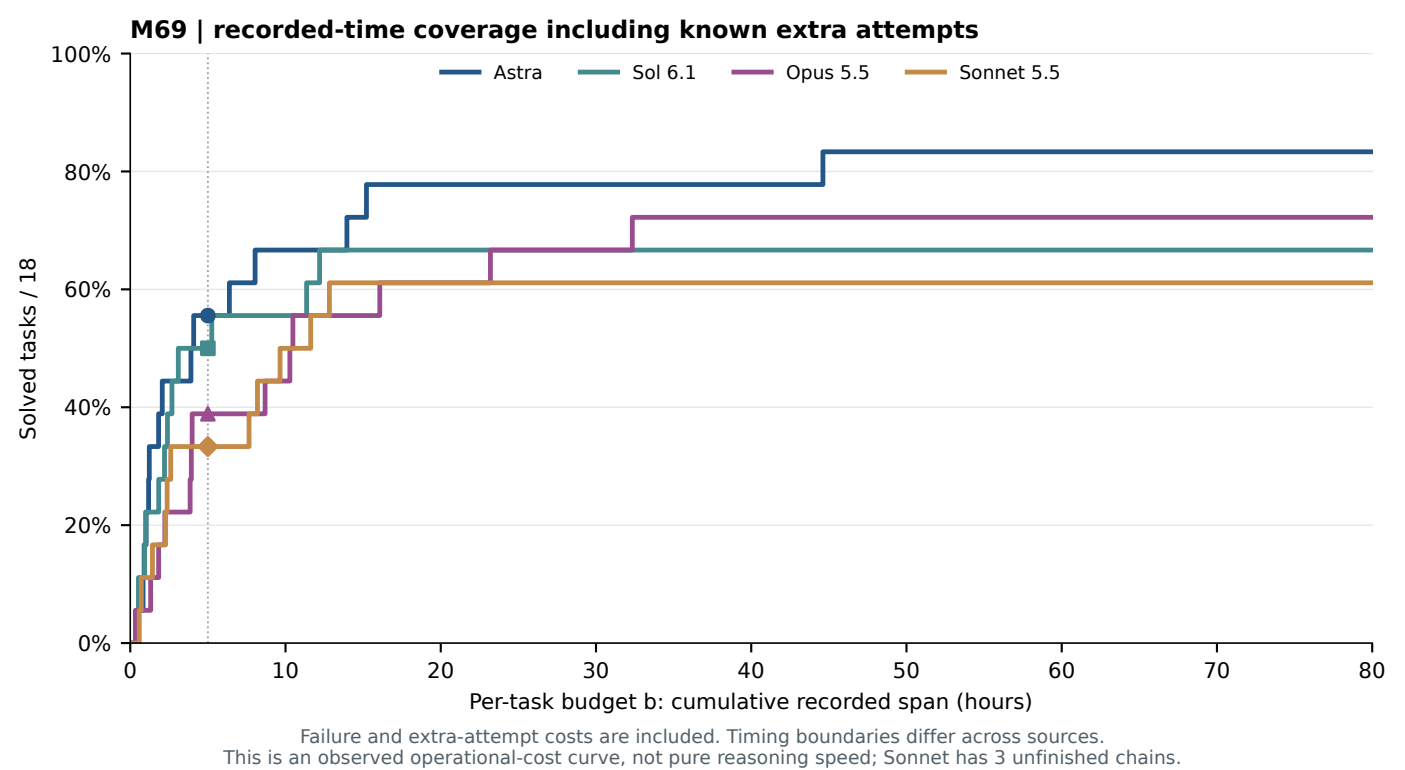
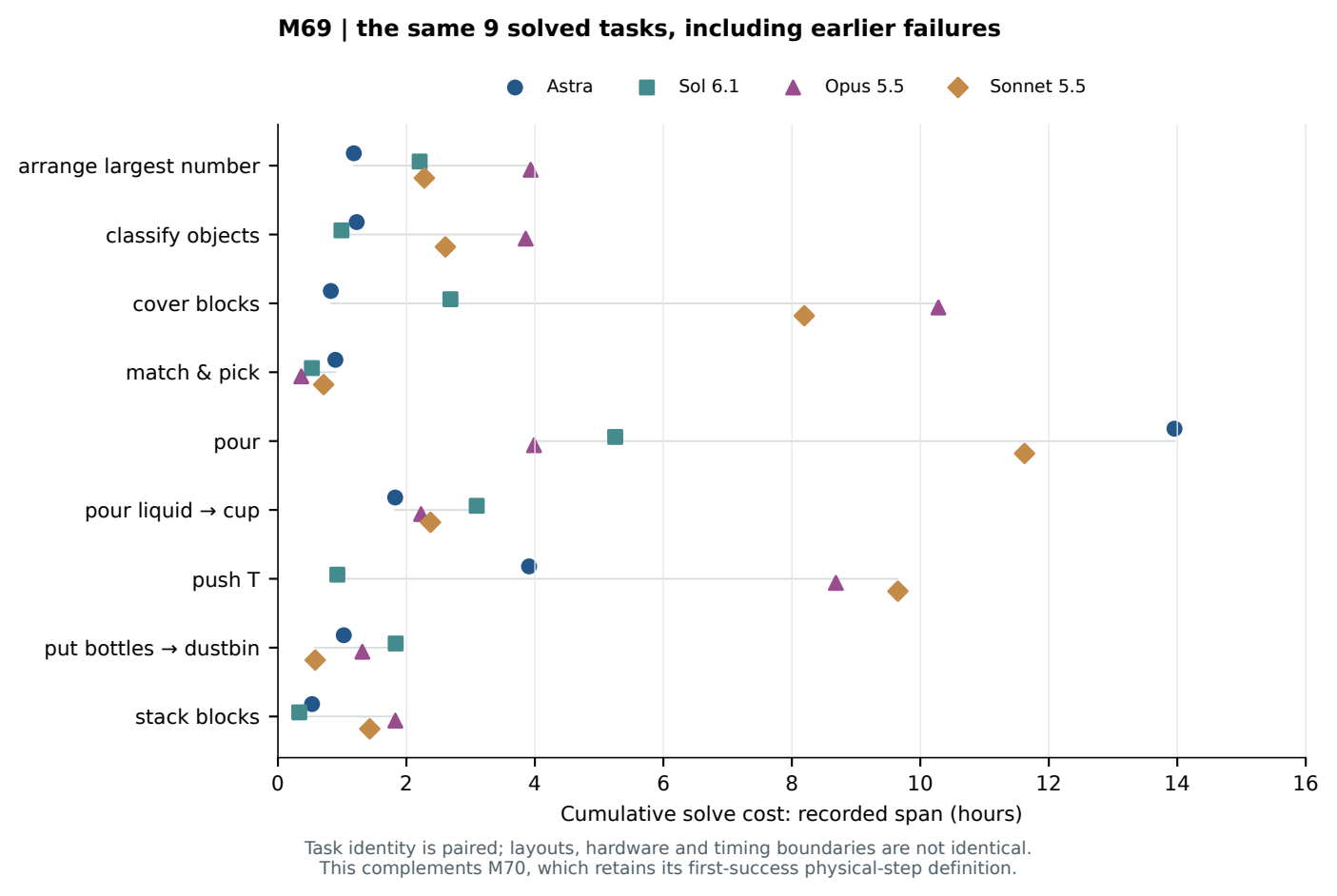


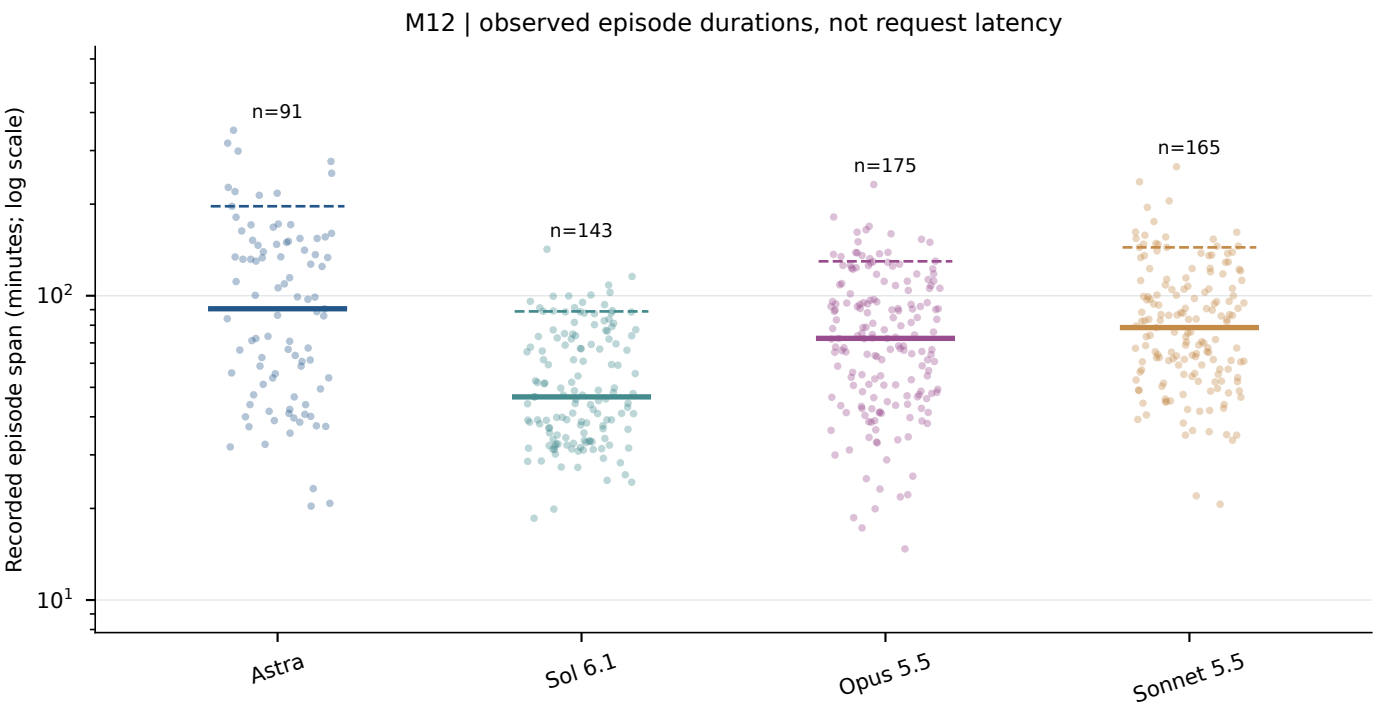
Canonical K1–K17 only. Astra: 71 measured + 20 estimated; Claude: TUI estimates.
Tokenizers and metering differ; these are usage records, not a monetary-cost ranking.



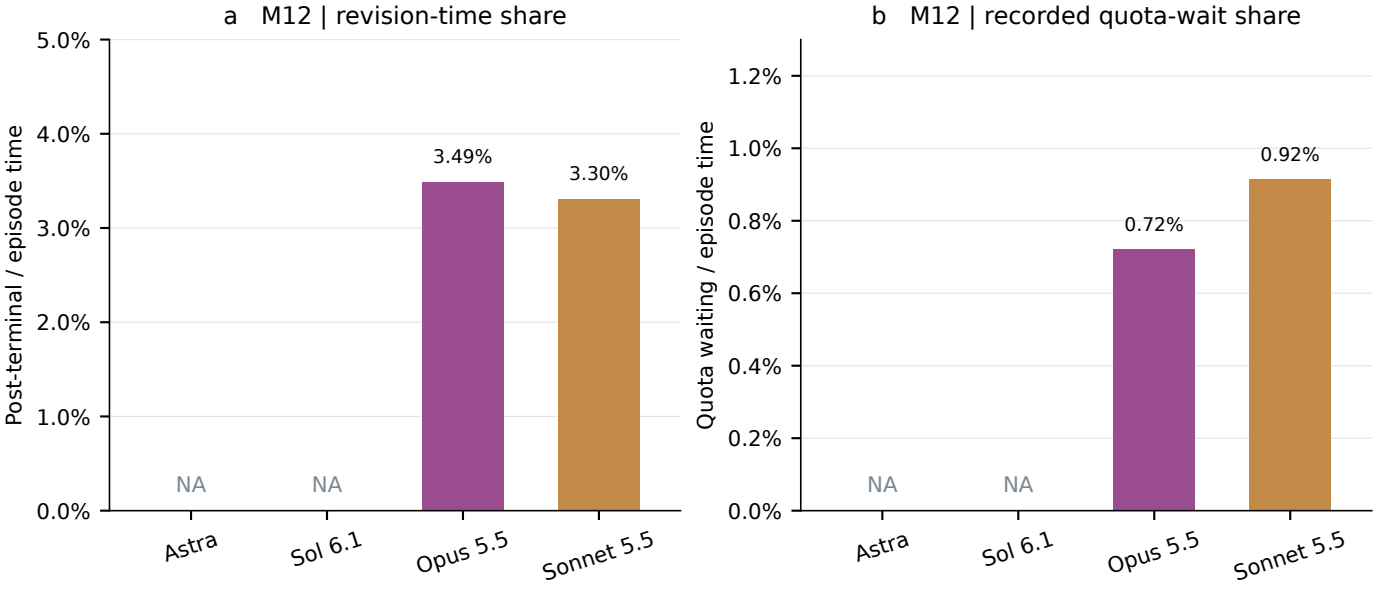
All known K1–K17 attempts with a span record; K0 excluded.
Episode durations are summed, including parallel runs. Unmetered directories remain excluded, not zero-cost.


M69 | recorded-time coverage including known extra attempts
Astra
Sol 6.1
Opus 5.5
Sonnet 5.5
Solved tasks / 18
100%
80%
60%
40%
20%
0%
0
10
20
30
40
50
60
70
80
Per-task budget b: cumulative recorded span (hours)
Failure and extra-attempt costs are included. Timing boundaries differ across sources.
This is an observed operational-cost curve, not pure reasoning speed; Sonnet has 3 unfinished chains.

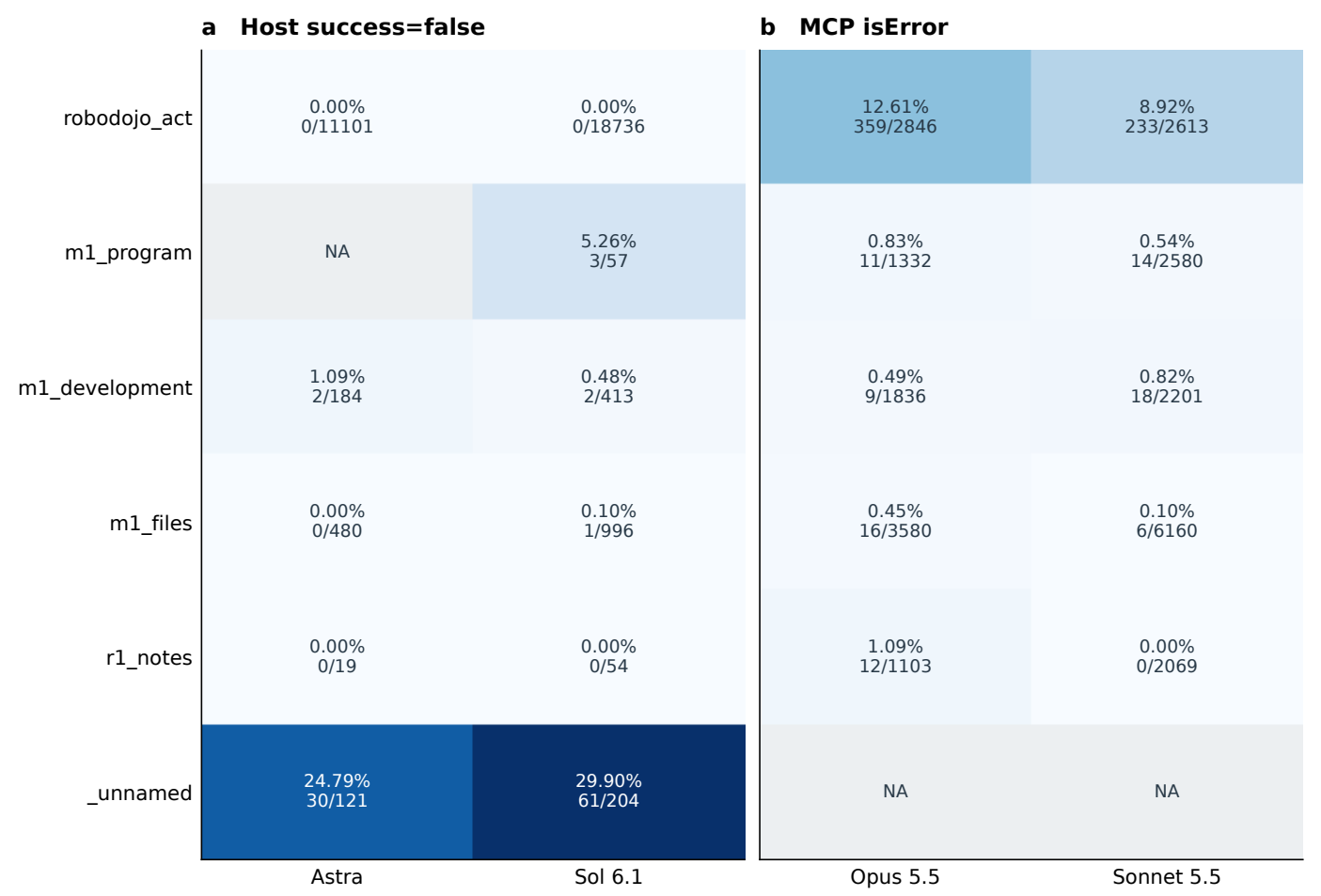

M69 | the same 9 solved tasks, including earlier failures
Astra
Sol 6.1
Opus 5.5
Sonnet 5.5
arrange largest number
classify objects
cover blocks
match & pick
pour
pour liquid → cup
push T
put bottles → dustbin
stack blocks
0
2
4
6
8
10
12
14
16
Cumulative solve cost: recorded span (hours)
Task identity is paired; layouts, hardware and timing boundaries are not identical.
This complements M70, which retains its first-success physical-step definition.



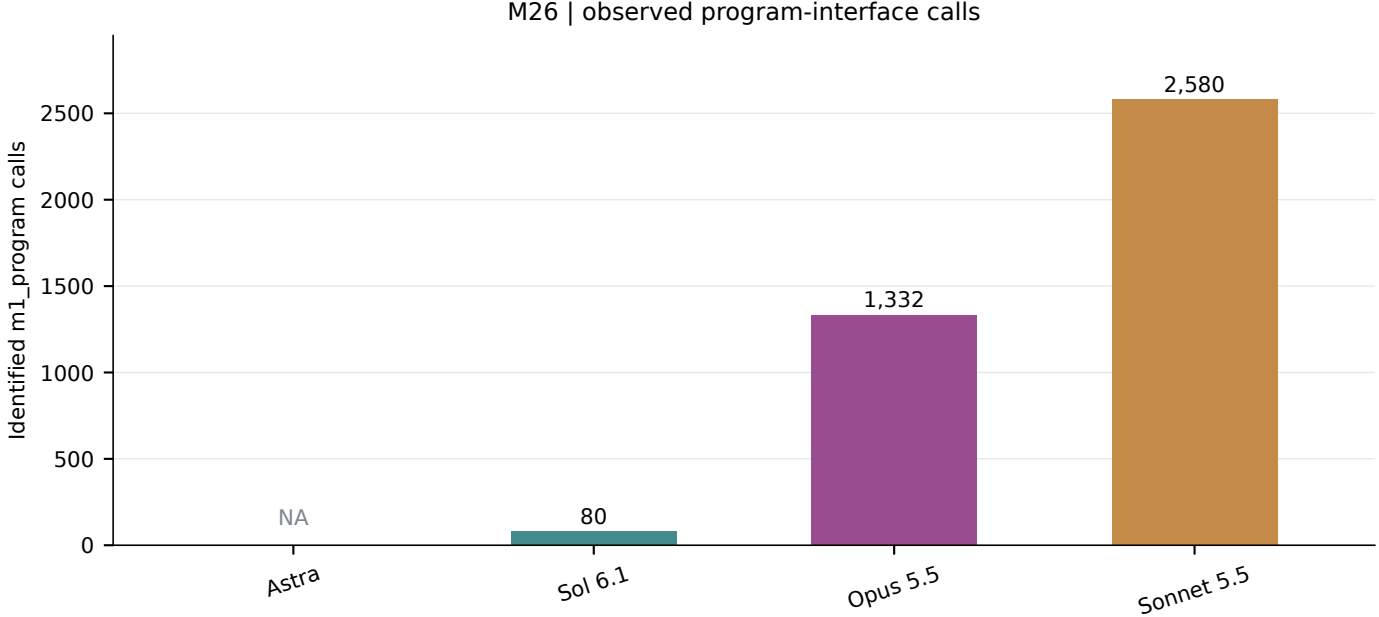
Each point is a canonical Harness episode. Solid line: median; dashed line: P90, not a confidence interval.
Astra mixes export spans and native-result fallback; other sources use report/session spans.

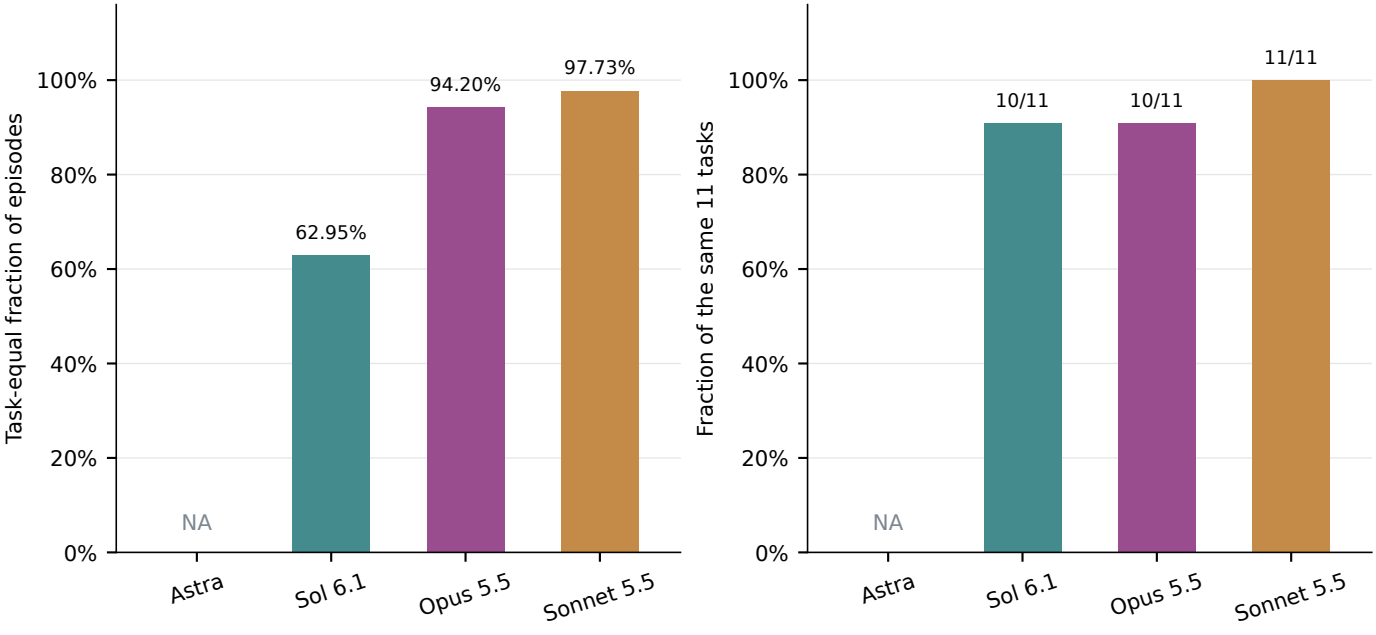


Revision boundaries: Opus 173/175; Sonnet 162/165. Last-event fallbacks excluded.
Quota waiting: canonical rounds only. NA means unavailable, not no revision or no waiting.



M24 | interface-error counts / calls, from the same logging layer.
Host and MCP error semantics differ; m1_program top-level errors are not a complete Python-exception rate.


M26 | observed program-interface calls
Identified m1_program calls
0
500
1000
1500
2000
2500
NA
80
1,332
2,580
Astra
Sol 6.1
Opus 5.5
Sonnet 5.5
Sol: literal parsing of exec code. Opus/Sonnet: directly logged MCP calls.
Call volume is a diagnostic, not a capability score; rejected direct actions invalidate simple action subtraction.



An access attempt means a recorded m1_development path pointing to an earlier round.
Astra has no matched canonical rollout. Access does not demonstrate successful reading or causal learning benefit.